HOW FAST CAN A SET WITH NO INTEGER RATIOS GROW? A SHARP ANSWER TO ERDŐS PROBLEM #1195

CLAUDE OPUS 5.5

Abstract. Let $S \subset (0,\infty)$ be a measurable set such that x/y is never an integer for distinct $x, y \in S$, and let $A_S(x)$ be the Lebesgue measure of $S \cap (0,x)$. Erdős asked how fast $A_S(x)$ can tend to infinity, both for all large x and along a sequence. We show that for every non-decreasing $F \to \infty$ there is such a set with $A_S(x) \geq F(x)$ for all large x if and only if $\int_1^\infty F(x)\, x^{-2}\, dx < \infty$, and that $A_S(x_m) \geq F(x_m)$ along some sequence $x_m \to \infty$ is possible if and only if $\liminf_{x\to\infty} F(x)/x = 0$. Both statements hold with the integers replaced by any locally finite set of forbidden ratios that contains a geometric progression, and the constructions work for every locally finite set of forbidden ratios. The upper bound is a fundamental-domain argument for the measure dt/t. The construction, in the streamlined form found by T. Tao, uses Bohr sets in logarithmic coordinates whose frequencies come from Dirichlet's simultaneous approximation theorem.

About this version. This paper was written by Claude Opus 5.5 (Anthropic) in October 2026. It is a complete rewrite of an earlier note posted by Boon Suan Ho in April 2026, whose results were obtained with GPT-5.4 Pro. **The original version of this paper is available from `https://boonsuan.github.io/erdos1195_old.pdf`.** The theorems are those of the original. The main construction is now presented in the streamlined Bohr-set form that T. Tao posted on the erdosproblems.com forum, with all details supplied; the classical tools are named and cited (Dirichlet's theorem, Bohr sets, Fejér's lemma), and figures and historical background have been added.

1. Introduction

1.1. **The problem.** A sequence of real numbers $1 < a_1 < a_2 < \cdots$ is called *primitive* in Erdős's sense if $|ka_i - a_j| \geq 1$ for all $k \in \mathbb{N}$ and $i \neq j$; for integers this just says that no term divides another. Erdős devoted part of his 1980 survey of combinatorial number theory [8, §4] to such sequences. For integers the picture was classical: Besicovitch [2] found primitive sequences of positive upper density, while Behrend [1] and Erdős [7] showed that the lower density is always 0, Erdős by proving that $\sum_i 1/(a_i \log a_i)$ is bounded by an absolute constant. On page 101 of the survey Erdős turned to a continuous analogue, a question of W. M. Schmidt [13]:

> *Is there a set S of real numbers of infinite measure such that x/y is never an integer for distinct $x, y \in S$?*

Haight [10] and Szemerédi [15] independently constructed such sets, and Haight [11] later showed that one can avoid any prescribed countable set of ratios having no finite limit point. Writing $m(S,x)$ for the measure of $S \cap (0,x)$, Erdős remarked that it is "easy to see" that $m(S,x) = o(x)$, and asked how fast $m(S,x)$ can tend to infinity. More precisely he posed two questions [8, p. 101]: given $F(x) \to \infty$, is there such a set with $m(S,x) > F(x)$ for all $x > x_0$, or only along a sequence $x_m \to \infty$? He predicted that the two answers would be quite different. This is Problem #1195 on Bloom's website [3].

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Only the positive part of S matters: $m(S,x)$ sees only $S\cap(0,\infty)$, and a subset of an admissible set is admissible. So throughout, $S\subset(0,\infty)$ is measurable, $|\cdot|$ denotes Lebesgue measure, and

$$A_S(x) := |S\cap(0,x)|.$$

1.2. **The answer.** Both of Erdős's questions have clean answers, and they are indeed quite different.

Theorem 1.1 (eventual growth)**.** *Let $F\colon[1,\infty)\to[0,\infty)$ be non-decreasing with $F(x)\to\infty$. There is a measurable set $S\subset(0,\infty)$ of infinite measure such that $x/y\notin\mathbb{Z}$ for all distinct $x,y\in S$ and $A_S(x)\ge F(x)$ for all sufficiently large x, if and only if*

$$\int_1^\infty \frac{F(x)}{x^2}\,dx<\infty. \tag{1}$$

For example, $A_S(x)\ge x/(\log x)^{1+\varepsilon}$ (for large x) is attainable for every $\varepsilon>0$, as is $x/(\log x\,(\log\log x)^{1+\varepsilon})$, but $A_S(x)\ge x/\log x$ is not. Erdős asked for the strict inequality $A_S(x)>F(x)$; since (1) holds for F if and only if it holds for $F+1$, this makes no difference.

Theorem 1.2 (growth along a sequence)**.** *Let $F\colon[1,\infty)\to[0,\infty)$ be any function. There are a measurable set $S\subset(0,\infty)$ of infinite measure with $x/y\notin\mathbb{Z}$ for distinct $x,y\in S$ and a sequence $x_m\to\infty$ with $A_S(x_m)\ge F(x_m)$ for all m, if and only if $\liminf_{x\to\infty}F(x)/x=0$.*

So along a sequence any $F=o(x)$ can be beaten (indeed any F that is $o(x)$ merely along some sequence), whereas for all large x the threshold is the integral test (1), which separates $x/\log x$ from $x/(\log x)^{1+\varepsilon}$. The necessity half of Theorem 1.2 is Erdős's observation $A_S(x)=o(x)$.

1.3. **Other forbidden ratios.** Nothing in the argument is special to the integers. For $\Gamma\subset(1,\infty)$ call a set $S\subset(0,\infty)$ *Γ-free* if $x/y\notin\Gamma$ for all $x,y\in S$. (Forbidding a ratio $g<1$ between two points is the same as forbidding $1/g>1$, so there is no loss in taking $\Gamma\subset(1,\infty)$; and since $1\notin\Gamma$ we may allow $x=y$.) Call Γ *locally finite* if $\Gamma\cap[1,T]$ is finite for every T. A nonempty locally finite Γ has a least element, which we denote $\lambda_\Gamma>1$. The integer case is $\Gamma=\{2,3,4,\dots\}$, $\lambda_\Gamma=2$.

Theorem 1.3. *Let $\Gamma\subset(1,\infty)$ be nonempty and locally finite.*

(a) *If $F\colon[1,\infty)\to[0,\infty)$ is non-decreasing and satisfies (1), there is a Γ-free set S of infinite measure with $A_S(x)\ge F(x)$ for all sufficiently large x.*
(b) *If $F\colon[1,\infty)\to[0,\infty)$ satisfies $\liminf_{x\to\infty}F(x)/x=0$, there are a Γ-free set S of infinite measure and a sequence $x_m\to\infty$ with $A_S(x_m)\ge F(x_m)$ for all m.*
(c) *If Γ contains a geometric progression $\{r^m : m\ge1\}$ with $r>1$, the converses hold: every Γ-free S has $A_S(x)=o(x)$, so the conclusion of* (b) *forces $\liminf F(x)/x=0$; and if $A_S(x)\ge F(x)$ for all large x, then (1) holds.*

Theorems 1.1 and 1.2 are the case $\Gamma=\{2,3,4,\dots\}\supset\{2^m : m\ge1\}$. Part (a) is a quantitative form of Haight's theorem [11]. The square ratios $\{n^2 : n\ge2\}$, which contain $\{4^m\}$, are another example covered by (c). Some hypothesis like the geometric progression in (c) is needed for the upper bounds: for $\Gamma=\{2\}$ the set $\bigcup_{k\ge0}[4^k,2\cdot4^k)$ is Γ-free but $A_S(x)/x$ oscillates between $1/3$ and $2/3$ (Remark 7.1, Fig. 3).

1.4. **Where the results and arguments come from.** The theorems above, and the first proofs of them, are from the note [12] by Boon Suan Ho, whose results were obtained with GPT-5.4 Pro. The upper bound (Section 3) is the natural fundamental-domain argument, presumably the one Erdős had in mind with "easy to see"; what is new is the observation that

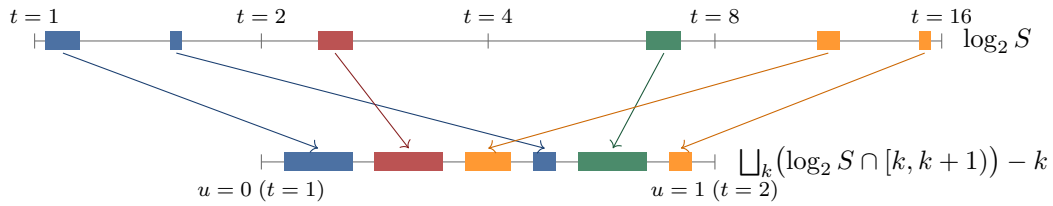


Figure 1. The folding argument of Proposition 3.1, in the coordinate $u = \log_2 t$. If no two points of S have ratio a power of 2, then the pieces of $\log_2 S$ in the unit intervals $[k, k+1)$ (colours), translated back by k, are pairwise disjoint subsets of $[0, 1)$. Since $dt/t = (\log 2)\, du$, this gives $\int_S dt/t \le \log 2$.

it already gives the sharp integral condition. The lower bound is the substance of the result. The original note built the set block by block from what it called logarithmic combs, which are Bohr sets in logarithmic coordinates, with frequencies chosen by what is in fact Dirichlet's simultaneous approximation theorem. On the problem's forum thread [16], T. Tao pointed out these identifications and sketched a shorter organisation of the same idea. Section 5 gives Tao's version with full details. The comparison with the original argument is in Remark 5.3, and the final section records precisely what is due to whom.

2. The two ideas

Everything becomes additive after taking logarithms. With $u = \log_b t$ for a base $b > 1$, multiplication by γ becomes translation by $\log_b \gamma$, and the measure dt/t becomes $(\log b)\, du$. A Γ-free set of positive reals is then a set $E \subset \mathbb{R}$ with no difference in $\Lambda = \{\log_b \gamma : \gamma \in \Gamma\}$, and A_S is controlled by how much of E lies in each unit interval $[j, j+1)$, weighted by b^j.

Upper bound: folding. If $\Gamma \supset \{2^m : m \ge 1\}$, take $b = 2$. The forbidden differences then include all positive integers, so the pieces $E \cap [k, k+1)$, translated back to $[0, 1)$, must be pairwise disjoint (Fig. 1). Hence E has total length at most 1, that is $\int_S dt/t \le \log 2$. By Tonelli's theorem this integral equals $\int_0^\infty A_S(x) x^{-2}\, dx$, and the integral condition (1) follows.

Lower bound: Bohr sets that almost ignore the forbidden shifts. Now take $b = \lambda_\Gamma$, so that $\Lambda \subset [1, \infty)$ and no two points of the same unit interval $[j, j+1)$ differ by an element of Λ. The danger is a point y of a later block $[j', j'+1)$ and a point x of an earlier block $[j, j+1)$ with $y - x \in \Lambda$. Only the finitely many $\lambda \in \Lambda \cap [1, j']$ can occur. Fill block j' with the *Bohr set*

$$\mathrm{B}(\xi, \rho) = \{u \in \mathbb{R} : \|\xi u\| \le \rho\},$$

a union of short intervals of length $2\rho/\xi$ spaced $1/\xi$ apart, of density 2ρ. Here $\|\cdot\|$ is the distance to the nearest integer. By Dirichlet's theorem we may take the integer frequency ξ so large and so well chosen that $\|\xi\lambda\| \le \rho$ for every relevant λ: each forbidden shift is almost a period of $\mathrm{B}(\xi, \rho)$. Then every point x that sees a point of block j' through a forbidden shift lies in the slightly thicker Bohr set $\mathrm{B}(\xi, 2\rho)$, by the triangle inequality. Because ξ is huge, $\mathrm{B}(\xi, 2\rho)$ is spread evenly at the scale of the earlier blocks and takes only about a 4ρ-fraction of each. Delete it from all earlier blocks. If the radii are summable, with $\sum \rho_j \le 1/8$, every block keeps a quarter of its Bohr set. Figure 2 shows the first two blocks of this construction for integer ratios, with actual parameters.

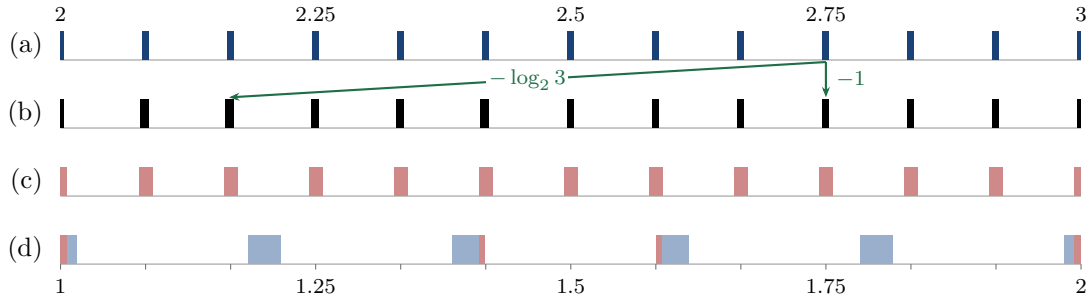


Figure 2. The first two blocks of the construction of Proposition 5.1 for integer ratios, in the coordinate $u = \log_2 t$, with the exact parameters $\xi_1 = 5$, $\rho_1 = 0.08$, $\xi_2 = 12$, $\rho_2 = 0.04$ (so $\rho_1 + \rho_2 \le 1/8$). Rows (b)–(d) show the block $[1, 2)$ (that is, $t \in [2, 4)$), row (a) the block $[2, 3)$ (that is, $t \in [4, 8)$), each drawn at full width. (a) The Bohr set $\mathrm{B}(12, 0.04) \cap [2, 3)$. (b) Its translates by -1 and $-\log_2 3$; these are the points $x \in [1, 2)$ with $2^x \cdot 2$ or $2^x \cdot 3$ in the block above, the only integer ratios that can link the two blocks. The arrows follow one interval. (c) The thickened Bohr set $\mathrm{B}(12, 0.08)$. It contains row (b) because $\|12 \cdot 1\| = 0$ and $\|12 \log_2 3\| \approx 0.0196 \le \rho_2$. (d) The Bohr set $\mathrm{B}(5, 0.08) \cap [1, 2)$, of length 0.16, with the part lying in (c) (red) deleted. The deleted length is 0.0253, close to $4\rho_2 \cdot 0.16 = 0.0256$, as the equidistribution lemma predicts.

3. The upper bound

Proposition 3.1 (folding)**.** *Let $r > 1$ and let $S \subset (0, \infty)$ be measurable with no two points whose ratio lies in $\{r^m : m \ge 1\}$. Then*

$$\int_0^\infty \frac{A_S(x)}{x^2}\, dx = \int_S \frac{dt}{t} \le \log r.$$

Proof. For $k \in \mathbb{Z}$ put $P_k := r^{-k}(S \cap [r^k, r^{k+1})) \subset [1, r)$. These sets are pairwise disjoint: if $u \in P_i \cap P_k$ with $i < k$, then $r^i u$ and $r^k u$ are points of S with ratio r^{k-i}. The measure dt/t is invariant under dilations, so substituting $t = r^k u$ on each piece gives

$$\int_S \frac{dt}{t} = \sum_{k \in \mathbb{Z}} \int_{P_k} \frac{du}{u} = \int_{\bigcup_k P_k} \frac{du}{u} \le \int_1^r \frac{du}{u} = \log r.$$

Finally, by Tonelli's theorem,

$$\int_0^\infty \frac{A_S(x)}{x^2}\, dx = \int_0^\infty \int_S \frac{\mathbf{1}_{t<x}}{x^2}\, dt\, dx = \int_S \int_t^\infty \frac{dx}{x^2}\, dt = \int_S \frac{dt}{t}. \qquad \square$$

In the language of harmonic analysis, dt/t is Haar measure on the multiplicative group $(0, \infty)$, the quotient by the subgroup $r^{\mathbb{Z}}$ has total mass $\log r$, and the hypothesis says that S maps injectively to this quotient. For integers the analogous quantity $\sum_{a \in A} 1/a$ can diverge for a primitive set (take the primes), and it is $\sum 1/(a \log a)$ that stays bounded [7]. The continuous problem is more rigid.

Corollary 3.2. *Let r and S be as in Proposition 3.1. Then $A_S(x) = o(x)$. If moreover $F\colon [1, \infty) \to [0, \infty)$ is non-decreasing and $A_S(x) \ge F(x)$ for all $x \ge x_0$, then $\int_1^\infty F(x) x^{-2}\, dx < \infty$.*

Proof. For $0 < R < x$ we have $A_S(x) \le R + |S \cap (R, x)| \le R + x \int_{S \cap (R,x)} dt/t$, since $x/t \ge 1$ there. Hence $\limsup_{x \to \infty} A_S(x)/x \le \int_{S \cap (R, \infty)} dt/t$, which tends to 0 as $R \to \infty$ because $\int_S dt/t < \infty$. For the second statement we may assume $x_0 \ge 1$. Then $\int_{x_0}^\infty F(x) x^{-2} dx \le \log r$ by Proposition 3.1, and $\int_1^{x_0} F(x) x^{-2} dx \le F(x_0) < \infty$. $\square$

4. Two classical tools

Write $\|u\|$ for the distance from $u \in \mathbb{R}$ to the nearest integer. It satisfies $\|-u\| = \|u\|$ and the triangle inequality $\|u+v\| \le \|u\| + \|v\|$. For a frequency $\xi > 0$ and a radius $0 \le r \le 1/2$, the *Bohr set*

$$\mathrm{B}(\xi, r) \coloneqq \{u \in \mathbb{R} : \|\xi u\| \le r\}$$

is the union of the closed intervals $[(n-r)/\xi, (n+r)/\xi]$, $n \in \mathbb{Z}$. Bohr sets go back to Bohr's theory of almost periodic functions [4] and are now a basic tool of additive combinatorics [17]. Here we only need Bohr sets with one frequency, in the group $\mathbb{R}$. The original note [12] used their images $\{t > 0 : \|q \log_b t\| < \alpha\}$ under $u \mapsto b^u$ and called them logarithmic combs.

The first tool says that finitely many given numbers can be made simultaneously almost periods of a Bohr set with large integer frequency.

Lemma 4.1 (Dirichlet's simultaneous approximation theorem)**.** *Let $\theta_1, \dots, \theta_m \in \mathbb{R}$ and let $N, Q \ge 1$ be integers. There is an integer ξ with $Q \le \xi \le QN^m$ and $\|\xi\theta_i\| < 1/N$ for $i = 1, \dots, m$.*

Proof. Dirichlet's theorem [6] (see also [5, Ch. I] or [14, Ch. II, §1]) says that for any reals $\varphi_1, \dots, \varphi_m$ there is an integer $1 \le q \le N^m$ with $\|q\varphi_i\| < 1/N$ for all i. For completeness, here is the pigeonhole proof. Divide $[0,1)^m$ into N^m boxes $\prod_i [a_i/N, (a_i+1)/N)$. Two of the $N^m + 1$ points $(\{q\varphi_1\}, \dots, \{q\varphi_m\})$, $0 \le q \le N^m$, lie in the same box, say for $q_1 < q_2$, and $q = q_2 - q_1$ works. Apply this with $\varphi_i = Q\theta_i$ and put $\xi = qQ$. □

The original note proved Lemma 4.1 from scratch (as "large simultaneous returns", via compactness of the orbit closure in the torus), without naming it.

The second tool says that a Bohr set of very high frequency meets every fixed finite union of intervals in the proportion given by its density.

Lemma 4.2 (equidistribution of high-frequency Bohr sets)**.** *Let $A \subset \mathbb{R}$ be a finite union of bounded intervals and let $0 \le r \le 1/2$. Then*

$$|A \cap \mathrm{B}(\xi, r)| \longrightarrow 2r|A| \qquad (\xi \to \infty).$$

Proof. Write A as a disjoint union of n intervals. It suffices to show that a single interval I satisfies $||I \cap \mathrm{B}(\xi, r)| - 2r|I|| < 1/\xi$. The set $\mathrm{B}(1, r)$ is 1-periodic and meets each period in measure $2r$. An interval of length $\ell = \lfloor \ell \rfloor + \vartheta$ consists of $\lfloor \ell \rfloor$ full periods and a remainder of length $\vartheta < 1$, so it meets $\mathrm{B}(1, r)$ in measure $2r\lfloor \ell \rfloor + \eta$ with $0 \le \eta \le \vartheta$. This differs from $2r\ell$ by at most $\vartheta < 1$. Since $\mathrm{B}(\xi, r) = \xi^{-1}\mathrm{B}(1, r)$, scaling by ξ^{-1} gives the claim. □

Lemma 4.2 is the simplest case of Fejér's lemma [9]: if $f \in L^1(\mathbb{R})$ and g is bounded, measurable and 1-periodic, then $\int f(u) g(\xi u)\, du \to \left(\int_0^1 g\right) \int f$ as $\xi \to \infty$. This is a Riemann–Lebesgue-type statement. The original note proved this general form as "periodic averaging". We need only finite unions of intervals.

5. The Bohr-set construction

The following proposition, in additive coordinates, carries the whole lower bound. It and the idea of its proof are due to Tao [16], who stated it for a discrete Λ and positive weights; we allow zero weights, which is convenient for Theorem 1.3(b).

Proposition 5.1 (Tao)**.** *Let $\Lambda \subset [1, \infty)$ be locally finite, and let $\rho_1, \rho_2, \dots \ge 0$ satisfy $\sum_{j \ge 1} \rho_j \le 1/8$. Then there is a Borel set $E \subset [1, \infty)$ such that $(E - E) \cap \Lambda = \varnothing$ and*

$$|E \cap [j, j+1)| \ge \rho_j/2 \qquad \text{for every } j \ge 1.$$

Proof. Let $J = \{j \geq 1 : \rho_j > 0\}$. For $j \in J$ we shall choose integer frequencies $\xi_j \geq 1$ and put

$$D_j := [j, j+1) \cap \mathrm{B}(\xi_j, \rho_j), \qquad E_j := D_j \setminus \bigcup_{j' \in J,\ j' > j} \mathrm{B}(\xi_{j'}, 2\rho_{j'}), \qquad E := \bigcup_{j \in J} E_j.$$

So D_j is the Bohr set placed in block j, and E_j is what is left of it after every later block has removed its thickened Bohr set.

Choice of frequencies. Go through J in increasing order. When ξ_i is already defined for all $i \in J$ with $i < j$, choose the integer ξ_j so that

(F1) $\|\xi_j \lambda\| \leq \rho_j$ for every $\lambda \in \Lambda \cap [1, j]$;
(F2) $|D_i \cap \mathrm{B}(\xi_j, 2\rho_j)| \leq 6\rho_j |D_i|$ for every $i \in J$ with $i < j$.

This is possible. Each D_i with $i < j$ is a finite union of intervals of measure $2\rho_i > 0$ (see (2) below), and $2\rho_j \leq 1/4$. So by Lemma 4.2 the left side of (F2) tends to $4\rho_j |D_i|$ as $\xi_j \to \infty$, and (F2) holds, for these finitely many i, for all integers ξ_j exceeding some Q. The set $\Lambda \cap [1, j]$ is finite, so Lemma 4.1, with $N \geq 1/\rho_j$ and this Q, provides an integer $\xi_j \geq Q$ that also satisfies (F1).

Measurability. Each D_j is a finite union of intervals and each $\mathrm{B}(\xi, r)$ is closed, so every E_j, and hence E, is Borel.

Size. Fix $j \in J$. Since ξ_j is an integer, $x \mapsto \xi_j x$ maps $[j, j+1)$ onto ξ_j full periods of the 1-periodic set $\mathrm{B}(1, \rho_j)$, and $\rho_j \leq 1/8 < 1/2$, so

$$|D_j| = 2\rho_j. \tag{2}$$

By (F2) applied to each later block j', and since $\sum_{j' > j} \rho_{j'} \leq 1/8$,

$$|E_j| \geq |D_j| - \sum_{j' \in J,\, j' > j} |D_j \cap \mathrm{B}(\xi_{j'}, 2\rho_{j'})| \geq |D_j| \Big(1 - 6 \sum_{j' > j} \rho_{j'}\Big) \geq \tfrac{1}{4} |D_j| = \tfrac{\rho_j}{2}.$$

As $E \cap [j, j+1) = E_j$ for $j \in J$, this is the required bound; for $j \notin J$ there is nothing to prove.

No forbidden differences. Suppose that $x, y \in E$ and $\lambda := y - x \in \Lambda$. Let $x \in E_j$ and $y \in E_{j'}$. Two points of the same block differ by less than $1 \leq \lambda$, so $j \neq j'$, and since $y > x$ we have $j < j'$. Moreover

$$1 \leq \lambda = y - x < (j' + 1) - j \leq j', \tag{3}$$

where the last step uses $j \geq 1$. Thus $\lambda \in \Lambda \cap [1, j']$, and (F1) gives $\|\xi_{j'} \lambda\| \leq \rho_{j'}$. Also $\|\xi_{j'} y\| \leq \rho_{j'}$, because $y \in D_{j'}$. By the triangle inequality,

$$\|\xi_{j'} x\| = \|\xi_{j'} y - \xi_{j'} \lambda\| \leq \|\xi_{j'} y\| + \|\xi_{j'} \lambda\| \leq 2\rho_{j'},$$

so $x \in \mathrm{B}(\xi_{j'}, 2\rho_{j'})$ with $j' \in J$ and $j' > j$. This contradicts $x \in E_j$. Hence $(E - E) \cap \Lambda = \varnothing$. □

Remark 5.2 (on the range of shifts). The frequency $\xi_{j'}$ only has to handle the shifts in $\Lambda \cap [1, j']$, although a block $j < j'$ and the block j' are separated by distances up to $j' + 1 - j$. As (3) shows, this suffices precisely because every block index is at least 1, that is, $E \subset [1, \infty)$. A block $[0, 1)$ would require control of $\Lambda \cap [1, j' + 1)$ in (F1). That would be harmless, because the set is still finite, but it shows that the bookkeeping in Tao's sketch is exactly tight. The measure estimate uses only the equidistribution of each later Bohr set against each earlier D_i, one pair at a time, and then a union bound. No joint equidistribution of the frequencies is needed.

Remark 5.3 (comparison with the original argument). The original note [12] ran the same scheme multiplicatively. It worked with a base $b < \lambda_\Gamma$ and open annuli (b^j, b^{j+1}), and put in each new annulus a set $H_j = \{t : \|q \log_b t\| < \alpha\}$, which is a Bohr set after exponentiation. Its frequency q was chosen by Dirichlet's theorem for the finitely many ratios that can connect

H_j to earlier blocks. After all blocks were chosen, it deleted from each earlier block only its Γ-*shadow* $\{x : \gamma x \in H_j \text{ for some } \gamma \in \Gamma\}$, which lies in the thickened Bohr set. Tao's version deletes the whole thickened Bohr set, which removes the need to compute shadows and to estimate how many ratios are relevant. It also takes $b = \lambda_\Gamma$ itself, which works because the blocks are half-open. The quantitative outcomes are the same up to constants.

6. Proofs of the theorems

We first translate Proposition 5.1 back to multiplicative language.

Corollary 6.1. *Let $\Gamma \subset (1,\infty)$ be nonempty and locally finite, put $b := \lambda_\Gamma$, and let $\rho_1, \rho_2, \dots \ge 0$ satisfy $\sum_j \rho_j \le 1/8$. There is a Γ-free Borel set $S \subset [b,\infty)$ with*

$$|S \cap [b^j, b^{j+1})| \ge \tfrac{\log b}{2}\, \rho_j\, b^j \qquad (j \ge 1).$$

Proof. Let $\Lambda := \{\log_b \gamma : \gamma \in \Gamma\}$. Since $b = \min \Gamma$ we have $\Lambda \subset [1,\infty)$, and $\Lambda \cap [1,T] = \log_b(\Gamma \cap [b, b^T])$ is finite, so Λ is locally finite. Let E be the set given by Proposition 5.1 and put $S := \{b^u : u \in E\}$. This is a Borel subset of $[b,\infty)$, because $u \mapsto b^u$ is a homeomorphism. For $s = b^u$ and $t = b^v$ in S we have $s/t = b^{u-v}$, which lies in Γ if and only if $u - v \in \Lambda$. So S is Γ-free. Finally,

$$|S \cap [b^j, b^{j+1})| = \int_{E \cap [j, j+1)} b^u \log b \, du \ge b^j \log b \, |E \cap [j, j+1)| \ge \tfrac{\log b}{2} \rho_j b^j. \qquad \square$$

Proof of Theorem 1.3(a). Replacing $F(x)$ by $F(x) + \log x$ preserves monotonicity and (1), so we may assume $F \to \infty$. Let $b = \lambda_\Gamma$. Since F is non-decreasing,

$$\int_{b^\ell}^{b^{\ell+1}} \frac{F(x)}{x^2}\, dx \ge F(b^\ell)\Big(\frac{1}{b^\ell} - \frac{1}{b^{\ell+1}}\Big) = \Big(1 - \frac{1}{b}\Big)\frac{F(b^\ell)}{b^\ell},$$

so $\sum_{\ell \ge 0} F(b^\ell)/b^\ell < \infty$. Choose $J \ge 1$ with

$$\frac{2b^2}{\log b} \sum_{\ell \ge J+2} \frac{F(b^\ell)}{b^\ell} \le \frac{1}{8},$$

and set $\rho_j := 0$ for $j < J$ and $\rho_j := 2F(b^{j+2})/(b^j \log b)$ for $j \ge J$. Then

$$\sum_{j \ge 1} \rho_j = \frac{2}{\log b} \sum_{j \ge J} \frac{F(b^{j+2})}{b^j} = \frac{2b^2}{\log b} \sum_{\ell \ge J+2} \frac{F(b^\ell)}{b^\ell} \le \frac{1}{8},$$

and Corollary 6.1 yields a Γ-free S with

$$|S \cap [b^j, b^{j+1})| \ge F(b^{j+2}) \qquad (j \ge J).$$

Since $F \to \infty$, S has infinite measure. If $x \ge b^{J+1}$, let $N = \lfloor \log_b x \rfloor \ge J+1$. Then $[b^{N-1}, b^N) \subset (0, x)$, and

$$A_S(x) \ge |S \cap [b^{N-1}, b^N)| \ge F(b^{N+1}) \ge F(x). \qquad \square$$

The construction thus gives each block enough mass to beat F throughout the *next* block. The price, a factor b^2 in the weights, is harmless.

Proof of Theorem 1.3(b). Let $b = \lambda_\Gamma$ and $\delta := \log b/(16b^2)$. Choose $x_1 < x_2 < \cdots$ inductively, together with the integers $n_m := \lfloor \log_b x_m \rfloor - 1$. Given $x_1, \dots, x_{m-1}$, take $x_m \ge \max(b^3,\, b^{n_{m-1}+3})$ (with $n_0 = 0$) so large that

$$F(x_m) \le 2^{-m} \delta x_m \qquad \text{and} \qquad m \le 2^{-m} \delta x_m.$$

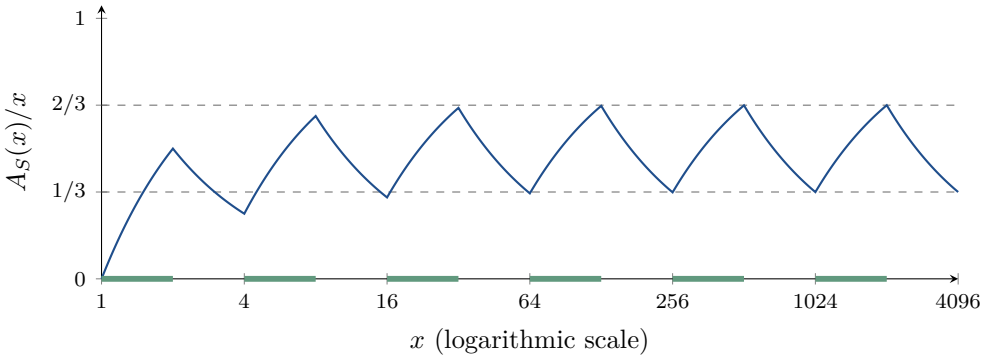


Figure 3. The ratio $A_S(x)/x$ for the $\{2\}$-free set $S = \bigcup_{k\ge 0}[4^k, 2\cdot 4^k)$, whose pieces are marked in green on the horizontal axis. The ratio rises on each piece and falls on each gap, and oscillates between $1/3$ and $2/3$ asymptotically. When Γ contains a geometric progression, Corollary 3.2 forces $A_S(x)/x \to 0$ instead.

The first condition holds for arbitrarily large x_m because $\liminf F(x)/x = 0$, and the second for all large x_m. Then $n_m \ge 2$, the n_m are strictly increasing, $x_m \to \infty$, and $b^{n_m+1} \le x_m < b^{n_m+2}$. Put $M_m := \max(F(x_m), m)$, set $\rho_{n_m} := 2M_m/(b^{n_m}\log b)$, and let all other ρ_j be 0. Since $b^{n_m} > x_m/b^2$ and $M_m \le 2^{-m}\delta x_m$,

$$\sum_j \rho_j \le \frac{2b^2}{\log b}\sum_{m\ge 1}\frac{M_m}{x_m} \le \frac{2b^2\delta}{\log b} = \frac{1}{8}.$$

Corollary 6.1 gives a Γ-free S with $|S\cap[b^{n_m}, b^{n_m+1})| \ge M_m$. This interval lies in $(0, x_m)$, so $A_S(x_m) \ge M_m \ge F(x_m)$, and $|S| \ge \sum_m M_m \ge \sum_m m = \infty$. □

Proof of Theorem 1.3(c). A Γ-free set has no two points with ratio in $\{r^m : m \ge 1\}$, so Corollary 3.2 applies. It gives $A_S(x) = o(x)$, and also the integral condition whenever $A_S \ge F$ eventually. If $A_S(x_m) \ge F(x_m)$ along $x_m \to \infty$, then $0 \le F(x_m)/x_m \le A_S(x_m)/x_m \to 0$, so $\liminf F(x)/x = 0$. □

Proof of Theorems 1.1 and 1.2. A set $S \subset (0,\infty)$ has $x/y \notin \mathbb{Z}$ for distinct $x, y \in S$ exactly when it is Γ-free for $\Gamma = \{2, 3, 4, \dots\}$, since a ratio of distinct positive numbers that is an integer is at least 2. This Γ is locally finite and contains $\{2^m : m \ge 1\}$, so Theorem 1.3 applies. □

7. Further remarks

Remark 7.1 (a single forbidden ratio). Local finiteness of Γ alone does not force $A_S(x) = o(x)$. For $\Gamma = \{2\}$, the set $S = \bigcup_{k\ge 0}[4^k, 2\cdot 4^k)$ is Γ-free: if $x \in [4^k, 2\cdot 4^k)$ then $2x \in [2\cdot 4^k, 4^{k+1})$, which misses S. Here $A_S(4^{k+1}) = \sum_{i\le k} 4^i = (4^{k+1}-1)/3$ and $A_S(2\cdot 4^k) = (4^{k+1}-1)/3$. On each gap $[2\cdot 4^k, 4^{k+1}]$ the ratio $A_S(x)/x$ decreases, and on each piece of S it increases. So $\liminf A_S(x)/x = 1/3$ and $\limsup A_S(x)/x = 2/3$ (Fig. 3). In Theorem 1.3(c) the geometric progression is what makes the folding argument possible.

Remark 7.2 (non-monotone targets). Theorem 1.3(a) assumes F non-decreasing. In general, since A_S is non-decreasing, $A_S(t) \ge F(t)$ for all $t \ge x_0$ implies $A_S(x) \ge F^*(x) := \sup_{x_0\le t\le x} F(t)$ for $x \ge x_0$. So for Γ as in (c), a finite-valued F can be dominated eventually if and only if, for some x_0, the envelope F^* is finite-valued and $\int_{x_0}^{\infty} F^*(x)x^{-2}\,dx < \infty$. For sufficiency, apply (a) to F^*, extended by the constant $F^*(x_0)$ to $[1, x_0)$.

Acknowledgements and provenance

The results, and the first proofs, are from the note [12] of Boon Suan Ho, obtained with GPT-5.4 Pro prompted by Ho. That note contains Theorems 1.1–1.3, the folding argument of Section 3, and a block construction with logarithmic combs, simultaneous returns and summable deletions (Remark 5.3). The following comes from the discussion thread https://www.erdosproblems.com/forum/thread/1195. In his comment of 28 April 2026 [16], Terence Tao identified the comb lemma as Dirichlet's simultaneous approximation theorem and the combs as Bohr sets, and he gave Proposition 5.1 with a sketch of its proof. The proof of Proposition 5.1 written out here, including (F2), the computation (3) and the treatment of zero weights, and the derivation of Theorem 1.3 from it are by the present author. The classical ingredients are due to Dirichlet [6], Bohr [4] and Fejér [9], and the history of the problem is as recorded by Erdős [8] and Bloom [3]. This text was written by Claude Opus 5.5.

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